

KPI Dictionary and Metric Definitions Framework

Data Analytics and Data Centre Operations
STC Solutions | 2018

Operating standard for consistent KPI definition, ownership, calculation, reporting and governance across managed data centre services.

Field	Details
Document owner	Teruyuki Ito
Operational scope	Data centre operations, service assurance, NOC reporting, capacity, change, asset and customer service reporting
Company context	STC Solutions - telecom and ICT managed services environment
Reporting period design	2018 operating model with daily, weekly and monthly reporting cadences
Document status	Internal working standard and metric governance reference
Version	v1.0

Positioning note

This framework is written as an operational artefact: it standardises how KPIs are defined, calculated, governed and consumed by service, operations, facilities, network, PMO and analytics stakeholders. It avoids ambiguous metric naming and creates a single reference point for reporting logic and ownership.

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1. Executive summary

This KPI dictionary was developed to create a consistent performance language across data analytics and data centre operations. Before the framework, operational reports could show different values for the same concept depending on source system, report owner, extraction timing or manual adjustment. The dictionary standardises KPI naming, calculation logic, owner accountability, refresh frequency, thresholds and escalation rules.

The framework is designed for a telecom and ICT managed services environment where data centre performance is measured across service availability, incidents, faults, capacity, changes, assets, customer reporting and data quality. It allows senior stakeholders to interpret performance at executive level while retaining enough diagnostic detail for NOC, facilities, network and service teams to take action.

- One agreed definition for each operational KPI, including business meaning, calculation logic and source-of-truth system.
- Clear separation between board-level KPIs, operational management KPIs and diagnostic metrics.
- Ownership model assigning each KPI to a business owner, data owner and reporting owner.
- Thresholds and exception rules to support weekly service reviews, monthly executive reporting and RCA governance.
- Metric certification process to reduce disputes over figures and improve confidence in dashboards.

2. Background and operating context

STC Solutions operated in a service environment where data centre operations, managed services, network operations, service assurance and customer reporting were closely linked. The KPI framework was therefore built to connect operational events to customer impact, service-level obligations and management decisions.

The operating model assumed that several teams were contributing to the KPI picture: NOC teams capturing alarms and faults, service teams managing incidents and escalations, facilities teams maintaining power and cooling resilience, network teams monitoring connectivity, change teams controlling release risk, and analytics teams consolidating the data into trusted reporting views.

Operational area	KPI responsibility within the framework
NOC / Fault Control	Alarm handling, first-line triage, incident creation, escalation and daily operational monitoring.
Service Assurance	SLA interpretation, customer impact validation, major incident coordination and service reporting.
Facilities Operations	Power, cooling, environmental monitoring, preventive maintenance and site resilience.
Network Operations	Core network availability, bandwidth utilisation, network fault isolation and service restoration.
Change / CAB	Change scheduling, risk review, customer notification, approval control and post-change validation.
Data Analytics / PMO	Metric definitions, dashboards, trend analysis, governance packs, RAID and action tracking.

3. KPI design principles

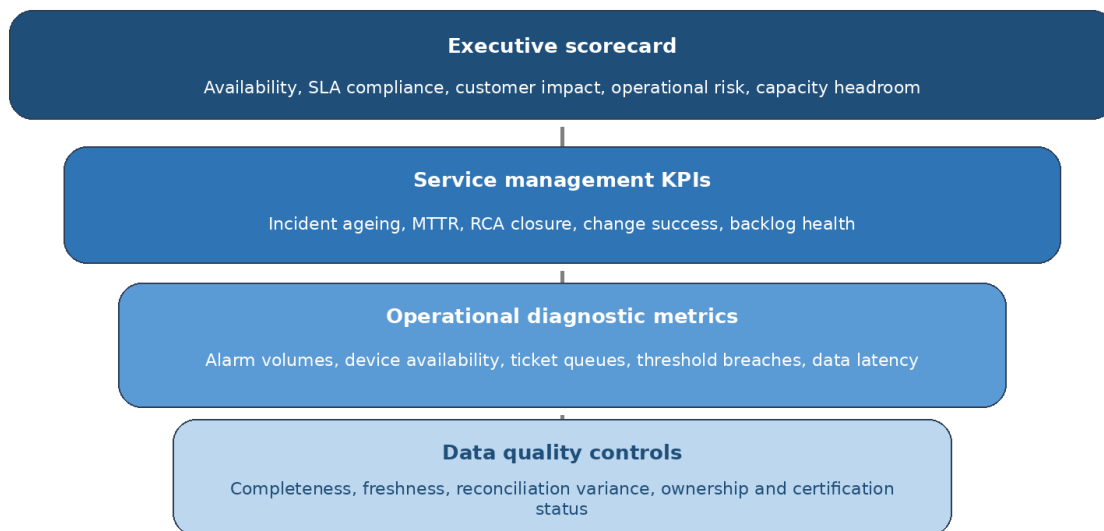
The KPI dictionary was based on practical reporting principles: metrics must be measurable, owned, repeatable and decision-oriented. A KPI was not added to the executive layer unless it had a confirmed business use, a stable calculation method and an accountable owner.

Principle	How it was applied
Single source of truth	Each KPI points to a nominated source system or certified reporting view. Where reconciliation is needed, the control rule is documented.
Clear business definition	The dictionary explains what the KPI means in operational language, not only in technical calculation terms.
Actionable thresholds	Green, amber and red criteria are linked to escalation, RCA, capacity planning or service review actions.
Traceability	Metrics can be traced from dashboard figure back to source records, filters and transformation logic.
Stable executive layer	Executive KPIs are kept stable month-to-month, while diagnostic metrics can evolve as operational risks change.
Ownership and accountability	Every metric has a business owner, data owner and reporting owner to avoid unmanaged figures.

4. KPI hierarchy and metric taxonomy

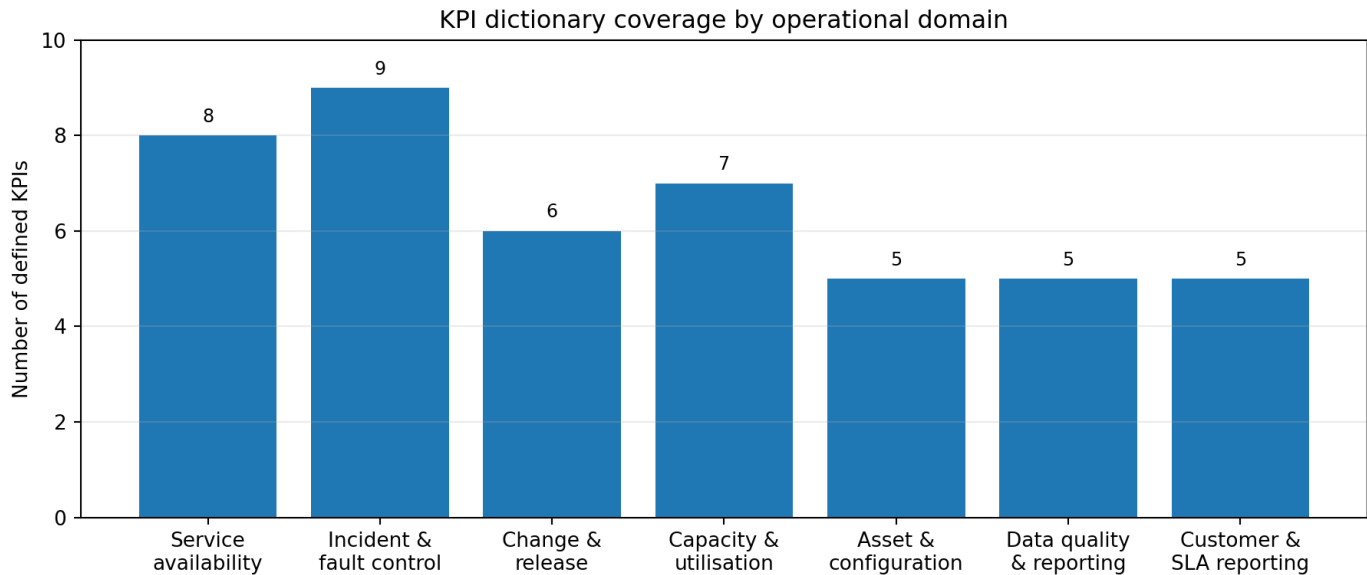
The framework separated KPIs into three practical layers. This avoided mixing executive reporting with operational troubleshooting. Senior leaders were given the small number of stable KPIs needed for governance, while operational teams retained drill-down metrics for root cause and corrective action.

KPI hierarchy used for data centre operations reporting



Design principle: board-level KPIs remain stable while operational and diagnostic layers provide drill-down for fault isolation and action ownership.

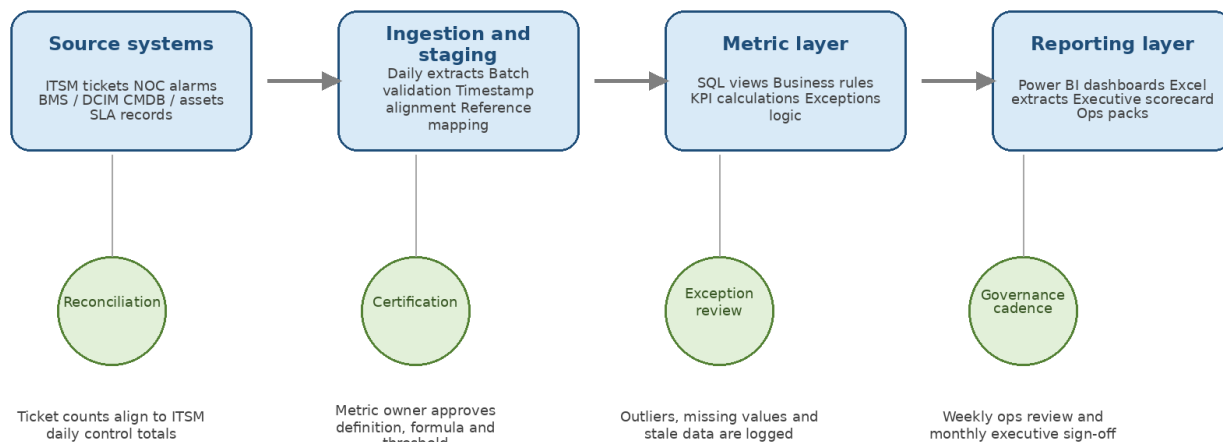
KPI domain	Business purpose
Service availability	Service continuity, SLA exposure, downtime and customer-impacting events.
Incident and fault control	Incident volume, severity, response, resolution, backlog, recurrence and RCA closure.
Change and release control	Change success, emergency change ratio, CAB controls and validation discipline.
Capacity and utilisation	Rack, power, cooling, bandwidth, threshold breaches and headroom forecast.
Asset and configuration	CMDB completeness, asset accuracy, lifecycle risk and monitoring coverage.
Data quality and reporting	Completeness, freshness, reconciliation, certification and dashboard usage.
Customer and SLA reporting	Customer report delivery, escalations, action closure and notification compliance.



5. Data flow, ownership and governance controls

The reporting design used a controlled flow from source systems into a curated metric layer. The aim was not only to visualise data, but to govern it. Every dashboard figure had to pass through defined rules for source extraction, transformation, reconciliation and ownership approval.

Data flow for KPI certification and operational reporting



Control outcome: each KPI has one approved definition, one owner, one refresh rule, and one documented escalation path.

Role	Responsibility
Business owner	Confirms the operational meaning of the KPI and agrees threshold rules.
Data owner	Confirms source availability, data definitions, refresh timings and field-level logic.
Reporting owner	Builds and maintains the certified report, dashboard or extract.
Control owner	Reviews exceptions, reconciliation failures and metric disputes.
Executive sponsor	Uses KPI outputs for governance decisions and removes blockers.

6. KPI dictionary and metric definitions

The following dictionary defines the core metrics used for data centre operations reporting. Each definition is structured so that operations, analytics and management stakeholders can interpret the same number in the same way. Where thresholds differ by customer contract, the documented contract target overrides the generic control rule.

6.0 Metric definition standard

Definition field	Standard applied
Metric name	Short business-friendly name used consistently in dashboards and governance packs.
Business definition	Plain-language explanation of what the metric means and why it matters operationally.
Calculation logic	Formula, numerator, denominator, inclusions, exclusions and approved time basis.
Primary source	System, table, extract or certified reporting view used as the source of truth.
Dimensions	Common drill-down fields such as site, service, customer, severity, owner, category and period.
Owner	Accountable role responsible for metric interpretation, dispute resolution and threshold approval.
Frequency	Refresh cadence and reporting cadence, including daily, weekly, monthly or quarterly use.
Control rule	Target, RAG threshold, exception rule or escalation trigger connected to management action.

Calculation convention

All percentages are reported as percentage points, not unit fractions. For time-based metrics, the start and end timestamps must be stated clearly. Where a customer contract defines a different SLA target, the contract-specific value overrides the generic threshold in this dictionary.

6.1 Service availability and SLA KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Service availability %	Percentage of contracted service time where the data centre service was available to customers.	$((\text{Total service minutes} - \text{customer-impacting outage minutes}) / \text{total service minutes}) \times 100$	ITSM outage log; NOC events; SLA calendar	Service Assurance	Daily / monthly	Green $\geq 99.90\%$; Amber $99.50\% - 99.89\%$; Red $< 99.50\%$
Power availability %	Availability of utility and backup power supporting production racks and critical facilities.	$((\text{Power service minutes} - \text{power-impacting outage minutes}) / \text{power service minutes}) \times 100$	BMS; UPS logs; generator test reports	Facilities Ops	Daily / monthly	Green $\geq 99.95\%$; any unplanned outage requires RCA
Network availability %	Availability of core network connectivity and customer-facing network services.	$((\text{Network service minutes} - \text{network-impacting downtime}) / \text{network service minutes}) \times 100$	NMS; router/switch monitoring; ITSM	Network Ops	Daily / monthly	Green $\geq 99.90\%$; P1/P2 events reviewed weekly
Cooling availability %	Availability of cooling systems serving live customer environments.	$((\text{Cooling service minutes} - \text{cooling-impacting downtime}) / \text{cooling service minutes}) \times 100$	BMS; DCIM; incident log	Facilities Ops	Daily / monthly	Green $\geq 99.90\%$; temperature breach triggers exception log
Unplanned downtime minutes	Total minutes of customer-impacting downtime excluding approved planned maintenance windows.	Sum of downtime minutes where <code>outage_type = unplanned</code> and <code>customer_impact = yes</code>	ITSM; outage bridge notes; SLA tracker	Service Assurance	Daily / monthly	Target: zero P1 downtime; trend reviewed by root cause category
Planned maintenance compliance %	Percentage of planned maintenance windows completed within approved schedule, scope and communication controls.	$(\text{Compliant planned maintenance windows} / \text{total planned maintenance windows}) \times 100$	Change calendar; maintenance schedule; customer notification log	Change Manager	Monthly	Green $\geq 95\%$; any customer-impacting variance logged
SLA compliance %	Percentage of service-level commitments met for availability, response and resolution.	$(\text{Number of SLA commitments met} / \text{total applicable SLA commitments}) \times 100$	SLA register; ITSM; customer contract matrix	Service Assurance	Weekly / monthly	Green $\geq 99\%$; Amber $97\% - 98.99\%$; Red $< 97\%$
Customer-impacting incidents	Count of incidents with confirmed customer service impact, regardless of severity.	Count incidents where <code>customer_impact = yes</code> within reporting period	ITSM incident table; escalation log	NOC Manager	Daily / weekly	Monitor week-on-week movement; all P1/P2 need RCA

6.2 Incident and fault control KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Incident volume	Total number of logged incidents by severity, site and service tower.	Count incident_id grouped by period, severity, site and category	ITSM incident table	NOC Manager	Daily / weekly	Investigate $> 15\%$ increase versus 4-week baseline
P1/P2 incident count	Number of critical and high-priority incidents opened in the reporting period.	Count incident_id where priority in (P1, P2)	ITSM incident table; major incident log	Major Incident Manager	Daily / weekly	P1: immediate escalation; P2: daily review until closure
Mean time to acknowledge (MTTA)	Average elapsed time between incident creation and first acknowledgement by support team.	$\text{Average}(\text{acknowledged_timestamp} - \text{created_timestamp})$	ITSM workflow timestamps	NOC Manager	Daily / weekly	Green ≤ 15 min P1/P2; exceptions reviewed daily
Mean time to resolve (MTTR)	Average elapsed time between incident creation and operational resolution.	$\text{Average}(\text{resolved_timestamp} - \text{created_timestamp})$, excluding paused customer/vendor waits where approved	ITSM workflow timestamps	Service Assurance	Weekly / monthly	Green $\leq$ target by severity; Red if $> 20\%$ over target
Open incident backlog	Number of incidents remaining open at period close.	Count incident_id where status not in (Resolved, Closed, Cancelled)	ITSM incident table	NOC Manager	Daily	Ageing buckets: 0-2d, 3-7d, 8-14d, $> 14d$
Backlog ageing > 14 days	Open incidents older than 14 calendar days without approved hold reason.	Count open incident_id where <code>age_days</code> > 14 and <code>hold_reason</code> is null	ITSM incident table; hold reason field	Service Assurance	Daily / weekly	Target $< 5\%$ of open backlog

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Incident reopen rate %	Percentage of resolved incidents reopened due to incomplete fix, recurrence or incorrect closure.	(Reopened incidents / resolved incidents) x 100	ITSM status history	Service Desk Lead	Weekly / monthly	Green <=3%; Red >5%
Repeat incident rate %	Percentage of incidents matching the same site/service/root cause within 30 days.	(Repeat incidents within 30 days / total incidents) x 100	ITSM; problem management tags; RCA register	Problem Manager	Monthly	Red if >10%; convert recurring patterns to problem record
RCA closure within SLA %	Percentage of required root cause analysis reports completed and approved within target time.	(RCAs closed on time / RCAs due) x 100	RCA register; major incident log	Problem Manager	Weekly / monthly	Green >=95%; P1 RCA due within agreed governance window

6.3 Change and release control KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Change success rate %	Percentage of approved changes implemented without rollback, incident or service degradation.	(Successful changes / total implemented changes) x 100	Change records; incident linkage; CAB notes	Change Manager	Weekly / monthly	Green >=97%; Amber 94%-96.99%; Red <94%
Failed change rate %	Percentage of implemented changes resulting in rollback, emergency fix or incident.	(Failed changes / total implemented changes) x 100	Change records; incident linkage	Change Manager	Weekly / monthly	Target <=3%; every failed change requires review
Emergency change ratio %	Share of emergency changes compared with all implemented changes.	(Emergency changes / total changes) x 100	Change calendar; CAB register	CAB Chair	Monthly	Green <=10%; Red >15% or upward trend for two months
Change collision count	Number of conflicting changes planned for same service, site or dependency window.	Count detected conflicts in approved change calendar	Change calendar; dependency register	PMO / Change	Weekly	Target zero unresolved conflicts before CAB approval
CAB approval lead time	Average number of days from change submission to CAB approval.	Average(CAB approval date - submitted date)	Change records; CAB minutes	Change Manager	Monthly	Monitor by change type and requestor team
Post-change validation completion %	Percentage of changes with completed technical and business validation evidence.	(Changes with completed validation / implemented changes) x 100	Change closure notes; validation checklist	Service Assurance	Weekly / monthly	Green >=98%; missing evidence aged and escalated

6.4 Capacity and utilisation KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Rack occupancy %	Percentage of available rack units allocated or occupied in each site/room.	(Occupied rack units / available rack units) x 100	DCIM; floor plan; asset allocation register	Capacity Manager	Weekly / monthly	Amber >=80%; Red >=90%; forecast expansion trigger
Power utilisation %	Actual power load as percentage of rated usable power capacity.	(Actual kW load / usable kW capacity) x 100	BMS; PDU readings; capacity model	Facilities Ops	Daily / weekly	Amber >=75%; Red >=85% sustained
Cooling utilisation %	Cooling load as percentage of usable cooling capacity, adjusted for redundancy requirements.	(Actual cooling load / usable cooling capacity) x 100	BMS; DCIM; thermal monitoring	Facilities Ops	Daily / weekly	Amber >=75%; Red >=85% or hotspot breach
Bandwidth utilisation %	Peak network throughput as percentage of provisioned bandwidth per service/customer segment.	(Peak throughput / provisioned bandwidth) x 100	NMS; bandwidth monitors; customer inventory	Network Ops	Daily / weekly	Amber >=70%; Red >=85% at peak for repeated days
Capacity headroom days	Estimated number of days before site/service reaches threshold based on recent trend.	(Threshold capacity - current utilisation) / average daily utilisation increase	DCIM; BMS; NMS; forecast model	Capacity Manager	Weekly / monthly	Red if <90 days; require capacity action plan
Threshold breach	Count of capacity, temperature,	Count threshold breach events	Monitoring tools; BMS;	Ops Tower	Daily /	Trend >10% requires

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
count	power or network thresholds breached in the period.	by type, site and severity	NMS	Leads	weekly	tuning or remediation
Forecast accuracy %	Accuracy of capacity forecast compared with actual utilisation after defined horizon.	$100 - \text{ABS}(\text{forecast} - \text{actual}) / \text{actual} \times 100$	Capacity forecast; actual utilisation extracts	Capacity Manager	Monthly	Green $\geq 90\%$; recalibrate assumptions if $< 85\%$

6.5 Asset, configuration and monitoring KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
CMDB completeness %	Percentage of required configuration items with mandatory fields populated.	$(\text{CIs with complete mandatory fields} / \text{total active CIs}) \times 100$	CMDB; asset discovery; site inventory	Configuration Manager	Weekly / monthly	Green $\geq 98\%$; Red $< 95\%$
Asset data accuracy %	Percentage of sampled asset records matching physical inventory and monitoring data.	$(\text{Accurate sampled records} / \text{sampled records}) \times 100$	CMDB; physical audit; monitoring inventory	Configuration Manager	Monthly / quarterly	Green $\geq 97\%$; mismatches logged as data quality actions
EOL/EOS exposure count	Number of assets or platforms past or approaching end-of-life/end-of-support.	Count active assets where EOL/EOS date $\leq$ reporting date + 180 days	Asset lifecycle register; vendor records	Asset Manager	Monthly	Red if customer-impacting asset has no remediation plan
Preventive maintenance compliance %	Percentage of scheduled preventive maintenance activities completed on time.	$(\text{PM tasks completed on time} / \text{PM tasks due}) \times 100$	Maintenance planner; work order system	Facilities Ops	Monthly	Green $\geq 95\%$; overdue critical tasks escalated weekly
Monitoring coverage %	Percentage of critical assets and services actively monitored with valid alert rules.	$(\text{Monitored critical CIs} / \text{total critical CIs}) \times 100$	Monitoring inventory; CMDB criticality flags	NOC Manager	Monthly	Green $\geq 98\%$; all P1 services require monitoring coverage

6.6 Data quality and reporting KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Data completeness %	Percentage of expected records and mandatory fields available for KPI calculation.	$(\text{Complete records} / \text{expected records}) \times 100$	Staging tables; DQ rules; source control totals	Data Analytics	Daily	Green $\geq 98\%$; Red $< 95\%$; dashboard warning applied
Data freshness SLA %	Percentage of scheduled data loads completed before agreed reporting cut-off.	$(\text{Loads completed on time} / \text{scheduled loads}) \times 100$	ETL logs; reporting refresh history	Data Engineering	Daily / weekly	Green $\geq 98\%$; Red if executive pack affected
Reconciliation variance %	Variance between reporting layer totals and source system control totals.	$\text{ABS}(\text{reporting total} - \text{source control total}) / \text{source control total} \times 100$	Source extracts; reporting mart; control log	Data Analytics	Daily / weekly	Green $\leq 1\%$; Red $> 3\%$
Metric certification status	Share of KPI definitions approved by accountable business and data owners.	$(\text{Certified KPIs} / \text{total KPIs in dictionary}) \times 100$	KPI dictionary; approval log	Data Governance	Monthly	Target 100% for executive KPIs
Dashboard adoption	Measured usage of standard dashboards by intended stakeholder groups.	Unique active users or views by dashboard / intended user population	BI audit logs; distribution lists	BI Lead	Monthly	Monitor usage drop; review if dashboard not used in two cycles

6.7 Customer and governance reporting KPIs

Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
Customer reporting SLA %	Percentage of customer reports delivered by committed reporting time.	$(\text{Reports delivered on time} / \text{reports due}) \times 100$	Reporting calendar; delivery log	Service Reporting	Weekly / monthly	Green $\geq 98\%$; all late reports require reason code
Escalation response time	Elapsed time from customer escalation receipt to first qualified response.	Average(first response timestamp - escalation received timestamp)	Escalation log; mailbox tracker; ITSM	Service Manager	Weekly / monthly	Measured by severity and customer segment
Executive action closure %	Percentage of agreed management actions closed by	$(\text{Actions closed on time} / \text{actions due}) \times 100$	Action log; governance minutes	PMO	Weekly / monthly	Green $\geq 90\%$; overdue actions escalated

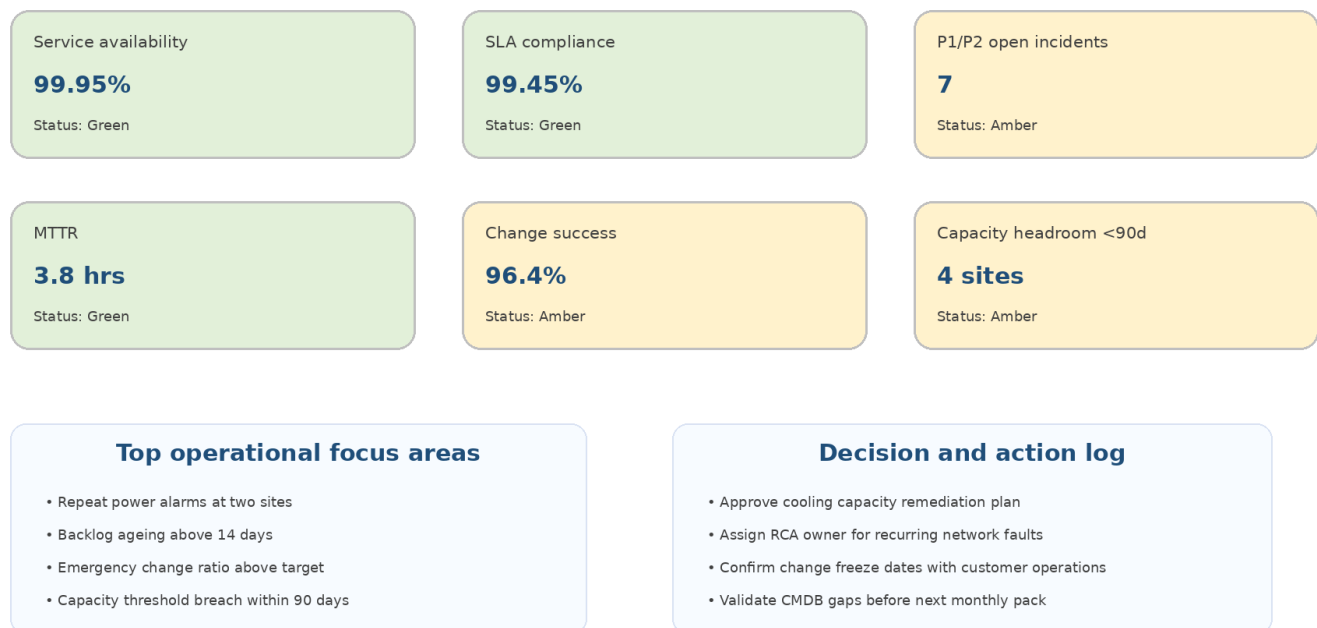
Metric / KPI	Definition	Calculation logic	Primary source	Owner	Frequency	Threshold / control rule
	target date.					
Repeat escalation count	Number of repeated customer escalations on same site, service or unresolved RCA theme.	Count repeated escalations with matching customer/site/theme within 30 days	Escalation log; RCA register	Service Manager	Monthly	Red if repeated twice within rolling 30 days
Customer-impacting change notification compliance %	Percentage of customer-impacting changes communicated within required notice period.	(Notified changes / customer-impacting changes) x 100	Change calendar; communication log; customer matrix	Change / Service Mgmt	Monthly	Green >=98%; non-compliance reviewed at CAB

7. Dashboard and reporting framework

The KPI dictionary was designed to feed a controlled reporting framework. Dashboards were intentionally split by audience to avoid overloading executive packs with diagnostic detail while still allowing operational users to drill into site, service, severity, owner, customer and time-period dimensions.

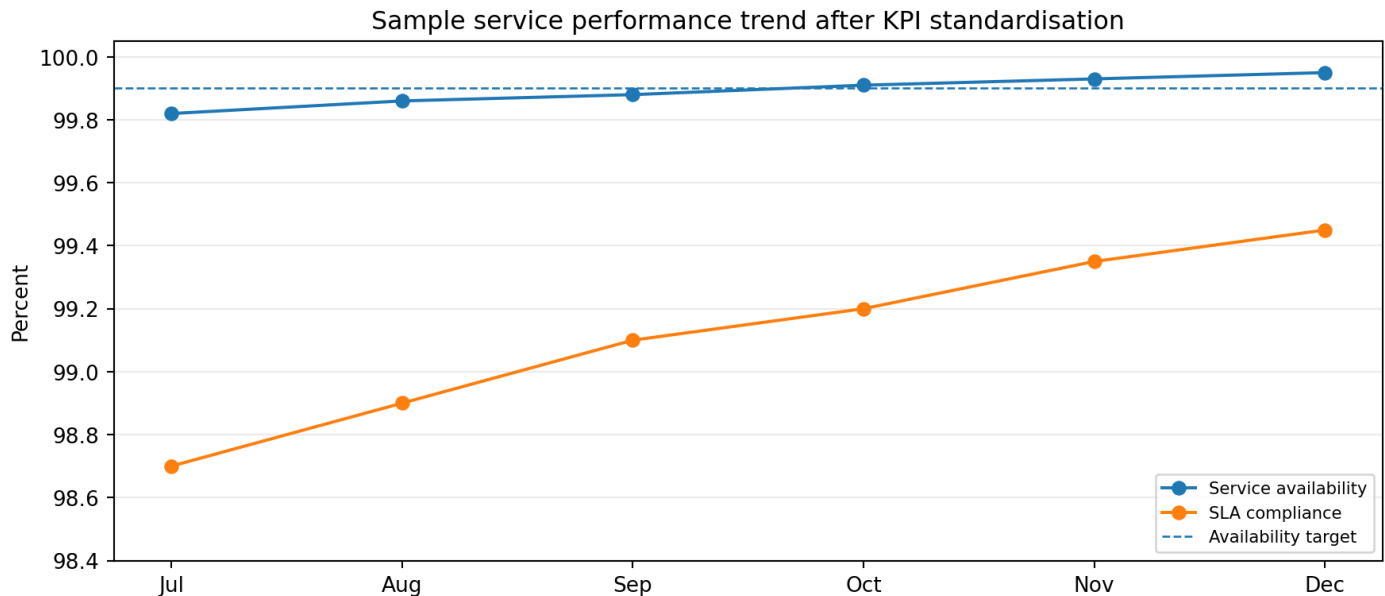
Report layer	Cadence	Main content	Primary audience
Executive scorecard	Monthly	Availability, SLA compliance, P1/P2 trend, capacity risk, customer escalations, overdue actions	Leadership / Service Governance
Weekly operations pack	Weekly	Incidents, backlog, MTTR, RCA, threshold breaches, change success, open risks	Ops leads / NOC / Facilities / Network
Daily fault control view	Daily	Open incidents, alarms, breached tickets, dispatch status, high-risk sites	NOC / Service Assurance
Capacity dashboard	Weekly / monthly	Rack, power, cooling, bandwidth, forecast headroom, threshold breach history	Capacity / Facilities / Network
Data quality monitor	Daily / weekly	Refresh status, missing data, reconciliation variance, certification status	Data Analytics / Data Engineering

Executive KPI scorecard layout - data centre operations



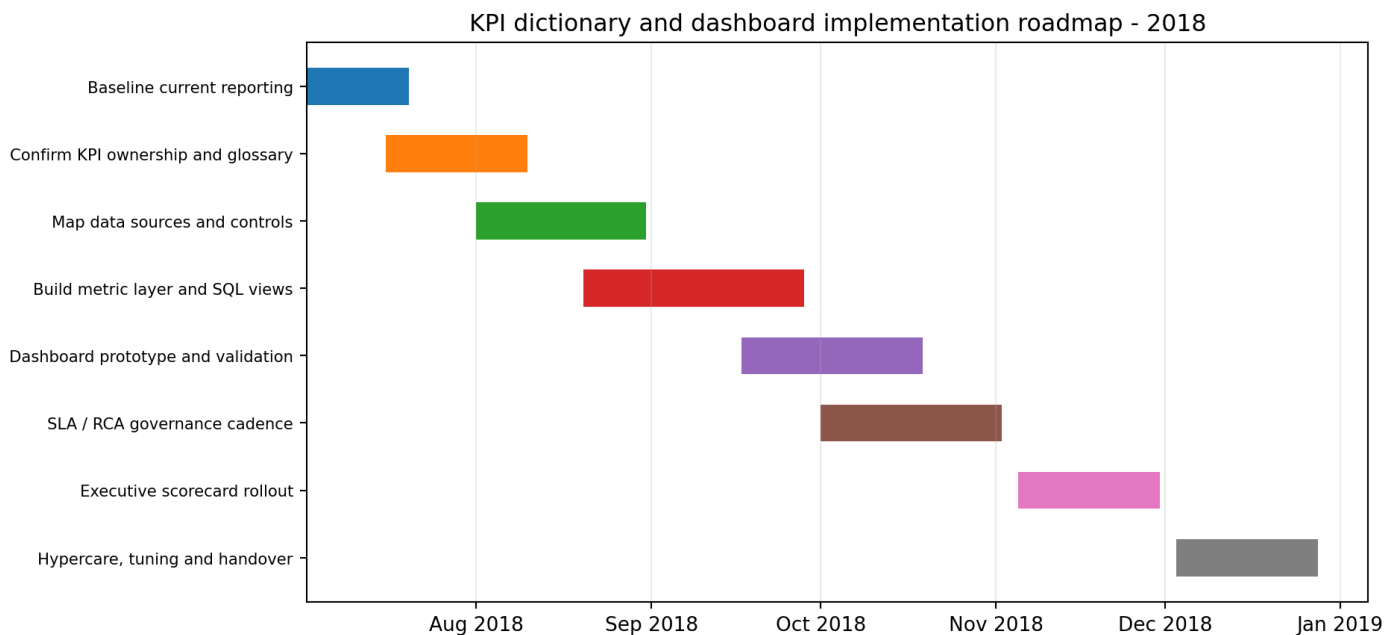
7.1 Dashboard page design

- Page 1 - Executive scorecard: headline status, RAG indicators, trend against target and open management decisions.
- Page 2 - Service availability: service uptime, unplanned downtime, customer impact, site/service breakdown and outage reason code.
- Page 3 - Incident and fault control: P1/P2 view, backlog ageing, MTTA, MTTR, reopen rate, repeat faults and RCA status.
- Page 4 - Capacity and resilience: rack, power, cooling and bandwidth utilisation, threshold breaches and forecast headroom.
- Page 5 - Change control: change volume, success rate, failed changes, emergency ratio, CAB lead time and validation gaps.
- Page 6 - Data quality: refresh status, completeness, reconciliation variance and metric certification progress.



8. Implementation roadmap and plan

The delivery plan was structured as a controlled governance rollout rather than a dashboard-only activity. The priority was to agree metric definitions first, then map data sources, build a metric layer, validate outputs with operational owners and finally roll the standardised scorecard into governance meetings.



Phase	Workstream	Activities	Output
Phase 1	Baseline and discovery	Inventory existing reports, document conflicting KPI definitions, confirm key users and pain points.	Current-state report catalogue; gap analysis
Phase 2	KPI definition and ownership	Agree KPI names, formulas, business meaning, owner, frequency and threshold rules.	Approved KPI dictionary v0.5
Phase 3	Data mapping and controls	Map source fields, extract timings, transformation rules, reconciliation controls and known data gaps.	Source-to-metric mapping; data quality checklist
Phase 4	Metric layer build	Create reusable SQL views or controlled data marts so all dashboards consume the same logic.	Certified metric layer; calculation test log
Phase 5	Dashboard prototype	Build executive, weekly ops and daily fault control views; validate trends with SMEs.	Prototype dashboard; UAT issue log
Phase 6	Governance rollout	Embed scorecards into weekly and monthly review	Live reporting pack; action log;

Phase	Workstream	Activities	Output
		cadence; assign action owners and exception rules.	hypercare tracker

9. Governance cadence and continuous improvement model

The KPI dictionary was maintained through a formal governance cadence. Metric changes were not made directly in dashboards without definition review, impact assessment and owner approval. This was important because even small changes to source filters, timestamps or exclusion rules could materially affect reported SLA and operational performance.

Cadence	Forum	KPI focus	Accountable roles
Daily	Fault control huddle	Open P1/P2, breached tickets, high-risk sites, stale data, customer-impacting events	NOC Manager / Service Assurance
Weekly	Operations performance review	Incident trend, backlog ageing, RCA, change success, capacity exceptions, DQ issues	Ops tower leads / PMO
Monthly	Executive service review	Availability, SLA, customer escalations, capacity risk, major decisions and overdue actions	Service leadership / executive sponsor
Quarterly	KPI dictionary review	Metric relevance, threshold changes, retired KPIs, new operational risks, data source changes	Data Analytics / Business owners
Ad hoc	Metric dispute or exception review	Conflicting figures, source outage, reconciliation variance or customer challenge	Metric owner / data owner / reporting owner

9.1 Metric change control

1. Submit a metric change request with reason, affected reports, affected stakeholders and expected impact.
2. Assess calculation impact using historical data, including before-and-after variance for at least three reporting periods where available.
3. Confirm approval from business owner, data owner and reporting owner before dashboard logic is updated.
4. Update the KPI dictionary, source-to-metric mapping and dashboard release notes.
5. Communicate change in the next governance pack so users understand why figures have moved.

10. Appendices

Appendix A - Standard RAG threshold method

Status	Meaning	Required action
Green	Metric is within target or tolerance and no immediate intervention is required.	Continue monitoring and note positive trend if sustained.
Amber	Metric is close to breach, trending negatively or outside normal operating range.	Assign owner, create action, monitor in next daily or weekly cycle.
Red	Metric has breached target, creates customer/SLA risk or requires management decision.	Escalate, document RCA or recovery plan, track until resolved.
Grey	Metric cannot be reported due to missing, stale or untrusted data.	Log data quality issue, apply warning on dashboard and assign data owner.

Appendix B - RACI for KPI governance

Activity	Business owner	Data owner	Reporting owner	Ops tower lead	Executive sponsor
Define KPI business meaning	A	C	C	I	I
Confirm data source and field mapping	C	A	R	I	I
Build dashboard or report view	C	C	A/R	I	I
Review weekly exceptions	A/R	C	R	C	I
Approve executive KPI changes	R	C	C	C	A
Resolve data quality issues	C	A/R	R	I	I

Legend: A = accountable, R = responsible, C = consulted, I = informed.

Appendix C - Data quality checklist

Control area	Validation question	Cadence
Completeness	Are all expected sites, services, tickets and time periods present?	Daily
Freshness	Did the source load complete before reporting cut-off?	Daily
Validity	Do fields contain accepted values for severity, site, status, owner and category?	Daily / weekly
Uniqueness	Are duplicate incidents, assets, work orders or alarms removed according to	Weekly

Control area	Validation question	Cadence
	agreed rule?	
Reconciliation	Do reporting totals tie back to source system control totals?	Daily / weekly
Lineage	Can the dashboard number be traced back to source fields, filters and calculation logic?	Monthly
Ownership	Does every KPI have an approved owner and escalation path?	Quarterly

Appendix D - KPI dictionary change log

Version	Date	Change summary	Owner
v0.1	Jul 2018	Initial report inventory and KPI consolidation structure created.	Data Analytics
v0.5	Aug 2018	Business definitions, ownership and primary formulas drafted with operations stakeholders.	Data Analytics / Ops Leads
v0.8	Oct 2018	Dashboard prototype validation completed; thresholds aligned to governance forums.	Service Assurance / PMO
v1.0	Dec 2018	Metric dictionary approved as internal working standard for operations reporting.	Data Analytics / Service Governance

Operating discipline

A KPI is only trusted when the definition, calculation, source, owner and decision path are all visible. The purpose of the dictionary is to make operational performance explainable, repeatable and actionable.